Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. **Team name:** Sarvam
b. **Team leader name:** Vigneswaran
c. **Problem Statement:** Official letters, bills, prescriptions and reports hide what to do and by when, in wording many people find hard to read. Sarvam explains any document in 9 Indian languages, and marks whether each fact matches the page.

Sarvam.

Team member: Abraham Donald
Theme: Sustainability & Social Impact
Live: sarvam-510715.web.app

Brief about the idea

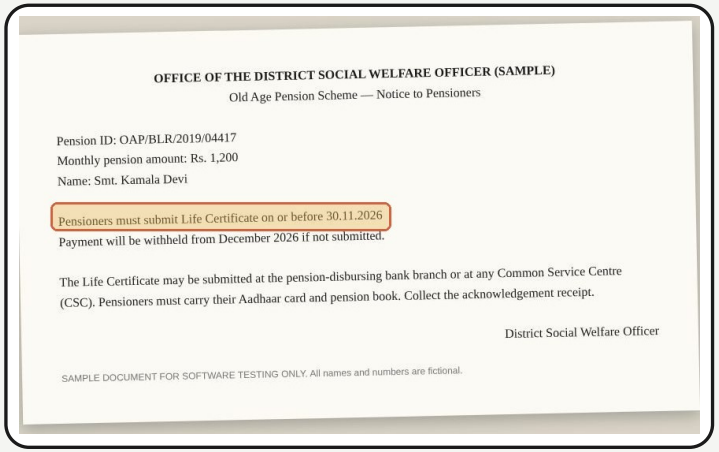
Any official document, explained in your language.

The problem. Notices, bills, lab reports and prescriptions hide what to do and by when, in wording many people find hard to read.

What Sarvam does. Photograph or upload the document. Sarvam says what it is, what to do and by when, in 9 Indian languages, out loud if you want.

Who it is for. Older people, first-time readers of official papers, and the families who help them.

1 The notice (fictional sample)



OFFICE OF THE DISTRICT SOCIAL WELFARE OFFICER (SAMPLE)

Old Age Pension Scheme — Notice to Pensioners

Pension ID: OAP/BLR/2019/04417
Monthly pension amount: Rs. 1,200
Name: Smt. Kamala Devi

Pensioners must submit Life Certificate on or before 30.11.2026
Payment will be withheld from December 2026 if not submitted.

The Life Certificate may be submitted at the pension-disbursing bank branch or at any Common Service Centre (CSC). Pensioners must carry their Aadhaar card and pension book. Collect the acknowledgement receipt.

District Social Welfare Officer

SAMPLE DOCUMENT FOR SOFTWARE TESTING ONLY. All names and numbers are fictional.

2 In Tamil, deadline first



3 The reminder

Submit Life Certificate
30 Nov 2026 · added to the calendar, alert 1 day before

Opportunities

Every fact says where it came from.

How different is it from any of the other existing ideas?

Chatbots answer and translation apps swap words. Sarvam turns a document into dated actions, picks a layout for its kind, and labels where each fact came from.

How will it be able to solve the problem?

Gemini reads the document. Our code checks each quote and number against it, and works out dates from the document's own rule.

USP of the proposed solution

Plain next steps in the reader's own language, each with an evidence label they can check.

WHAT SARVAM SAYS, AND WHERE IT CAME FROM

Matched

₹31,200 of the ₹48,500 you claimed is approved.

"Amount approved for payment: Rs. 31,200" Insurance letter PDF, p. 1

Calculated

Ask for a review by 31 Oct 2026.

"within 30 days of this letter", dated 01 October 2026 Insurance letter, p. 1

Check against original

Submit your Life Certificate by 30 Nov 2026.

"on or before 30.11.2026" Pension notice, photo

A label says where a fact came from. It is a check on that fact, not on the whole explanation.

List of features offered by the solution

One upload: actions, dates, answers and reminders.

Photo, PDF or pages
Many photos, read as one

23 document kinds
Bills, notices, reports, legal, money

Splits big PDFs
Checkup: report, prescription, bill

Lab flags as printed
High or Low only where printed

Medicine reminders
Doses and times as written

Deadline reminders
Future dates to your calendar

Listen in 8 languages
Natural voices, three speeds

Ask Sarvam
Text or voice; quotes the document

Hide personal details
On by default, any language

Save as image
Ready for WhatsApp or family

Easier to read
Larger text, contrast, simple words

Private by design
Kept on the phone, not on our server

Process flow diagram or Use-case diagram

From a photo to a plan: a median of 17 seconds.

1 — Choose language
9 languages or Auto

→

2 — Photo or PDF
One or many pages

→

3 — Gemini reads
Kind of document, facts, actions, in your language

→

4 — Code checks
Quotes, numbers and dates against the page

→

5 — Result in its layout
Next up first, then simple words and watch-outs

English uploads: median 17 s (n=7); 1-page PDF 13 s. Samples open in 1.6 s.

THEN, FROM THE RESULT

- Listen
- Ask by text or voice
- Share privately
- Save as image
- Add to calendar

YOU STAY IN CONTROL

You compare with the original
Anything read from a photo says “Check against original”.

You choose what to share
“Hide personal details” is on by default.

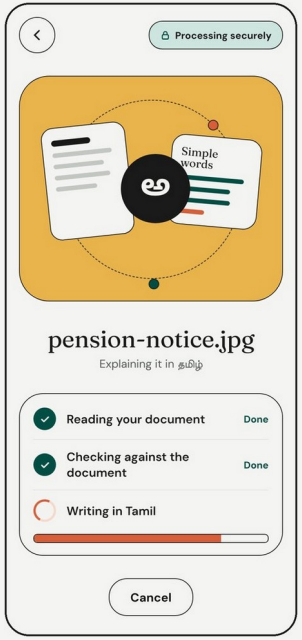
Sarvam explains, it does not advise
No medical, legal or financial advice.

Wireframes/Mock diagrams of the proposed solution (optional)

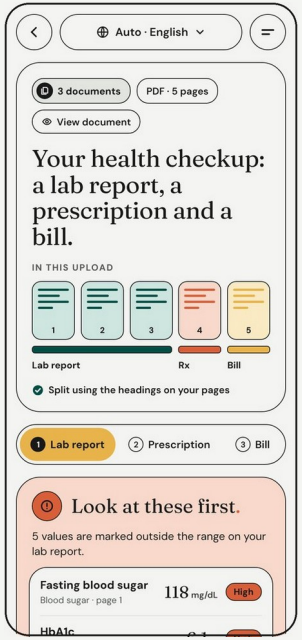
Designed for one thumb, big type and a calm pace.



Welcome in 9 scripts



Reading, with real progress



One PDF, three documents



Make Sarvam easier

DESIGN CHOICES

Language first: 9 scripts on every screen, Auto by default.

One main action per screen; touch targets of 44 pt or more.

Result order: what to do, simple words, watch out, then facts.

Larger text, high contrast and reduced motion settings.

TYPE AND COLOUR

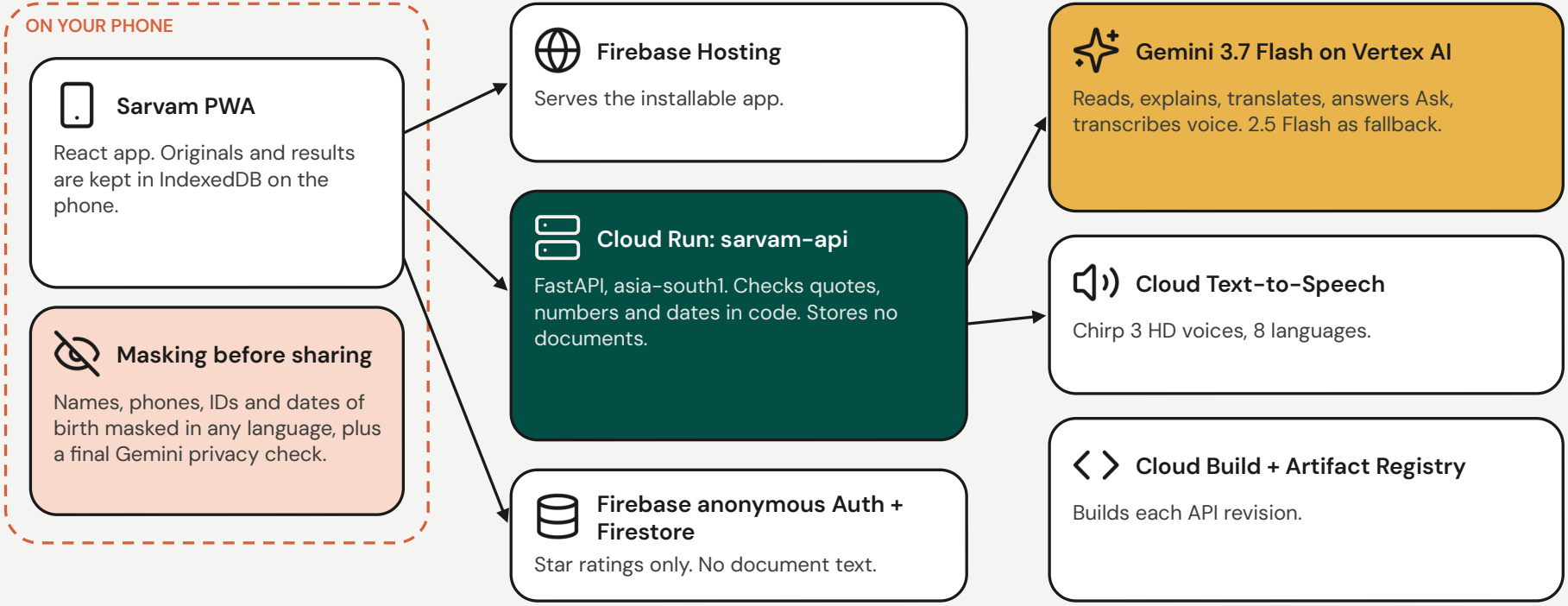
Fraunces + DM Sans + Noto Indic

Action Private Watch out Ask

Design mockups (design v3). Live screens are on slide 10.

Architecture diagram of the proposed solution

Read in the cloud, kept only on your phone.



Documents go to the API only to be read; our server keeps no copy. HTTPS only.

Technologies to be used in the solution

Google Cloud does the reading. Our code does the checking.

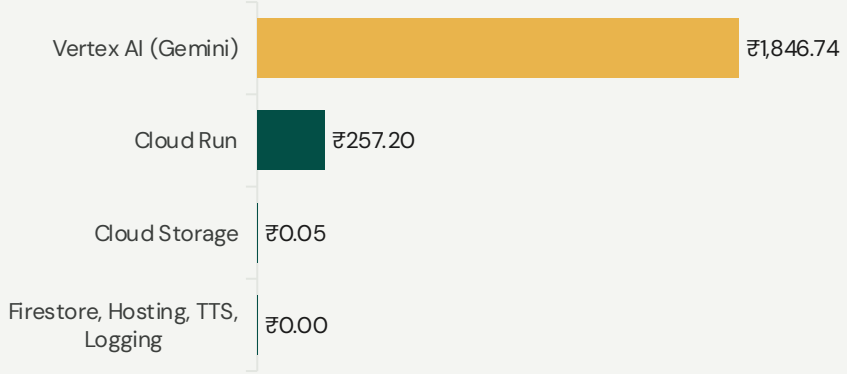
Gemini 3.7 Flash on Vertex AI	Reads, classifies, explains, translates, answers Ask, transcribes voice
Gemini 2.5 Flash on Vertex AI	Fallback model when the primary is busy, with the same prompts and checks
Cloud Run	Runs the FastAPI service: evidence matching, date calculation, rate limits
Firebase Hosting	Serves the installable PWA with security headers
Firebase anonymous Auth	Lets a reader rate a result without an account
Firestore	Stores star ratings only, never document text
Cloud Text-to-Speech	Listen in 8 Indian languages with Chirp 3 HD voices
Cloud Build + Artifact Registry	Container builds and revisions for Cloud Run

App stack: React, TypeScript, Vite, Tailwind, IndexedDB, service worker (PWA), Python, FastAPI, google-genai SDK.

Estimated implementation cost (optional)

₹2,104 of cloud usage in two weeks, ₹0 billed.

GOOGLE CLOUD BILLING, 30 SEP – 14 OCT 2026



₹2,103.99	−₹2,103.99	₹0.00
usage	discounts and trial credits	billed

Includes every development and QA test run. Trial credit left: ₹27,645.

WHERE THE MONEY WENT

Gemini 3.7 Flash (main model)	**₹1,753.84**
Gemini 2.5 Flash (fallback)	**₹92.88**
Cloud Run, one warm instance	**₹207.72**
Cloud Run, requests and network	**₹49.49**

3.7 Flash split: output ₹1,074, input ₹552, images ₹127.

HOW SARVAM KEEPS COST LOW (MEASURED)

≈ ₹0.94 per Gemini call. One call per explanation, written directly in your language.

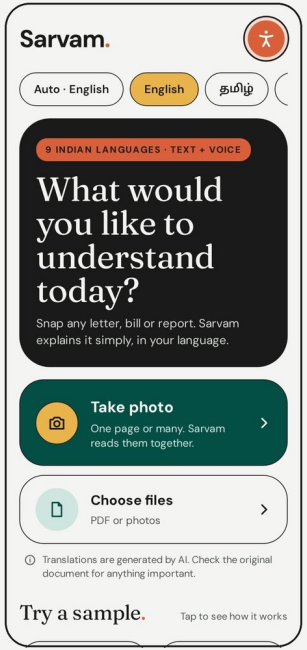
Same document again: 0.5 s and 0 AI calls, even after reopening the app (vs 23 s for a first read).

Samples: 11 documents pre-explained in 9 languages, so trying Sarvam costs ₹0.

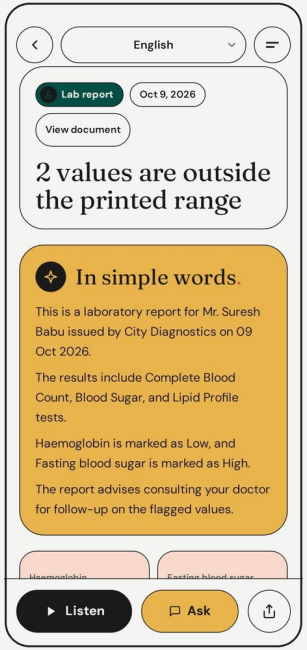
At scale: 10,000 new documents a month ≈ ₹10,000 + one warm Cloud Run instance; repeats are free. Hosting, Firestore, voice and logging stay in the free tier.

Snapshots of the prototype

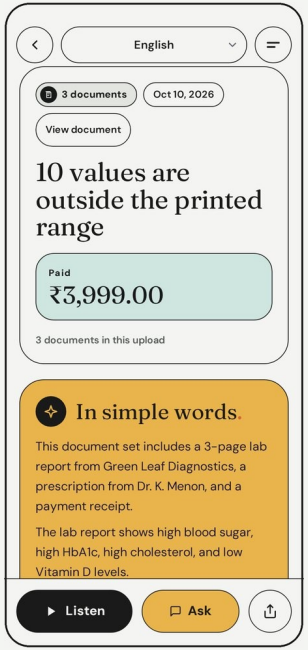
The running prototype, screen by screen.



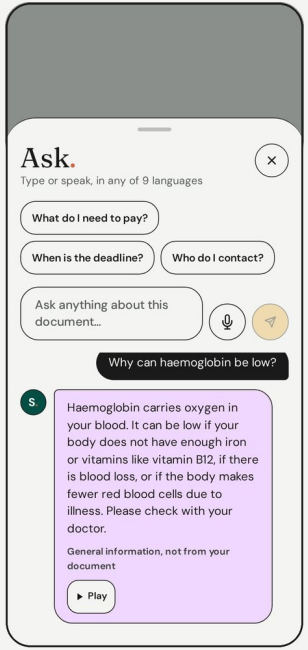
Home



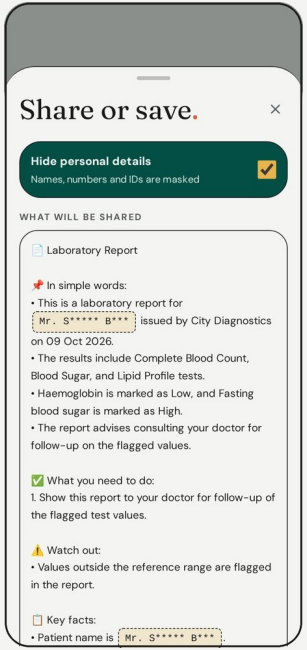
Lab report, flags as printed



One PDF, three documents



Ask: a general question



Share, details hidden

Prototype Performance report/Benchmarking

Measured on the running prototype, failures included.

16 / 17

uploads explained in the final check. The one miss: a Hindi PDF that timed out at 90 s.

17 s

median time to result, English uploads (n=7). 1-page PDF: 13 s.

211 / 211

expected facts correct across the 11 sample documents.

252

automated tests passing: 131 API and 121 web.

Measured with automated end-to-end tests on the deployed prototype, 14–15 Oct 2026, using 11 fictional sample documents and uploads in English, Hindi and Tamil. Small samples; counts shown.

Earlier full run: 66 of 68 explain calls succeeded; both failures were deliberately unreadable photos.

FINAL CHECK, 15 OCT 2026

Sample × language results that open	**96 / 99**
Hindi / Tamil upload, median	**15.6 s / 17.0 s**
5-page PDF / photo upload	**45 s / 71 s**
Ask reply, median (n=3)	**7.5 s**
Personal details leaked, Hide ON (96 samples + 4 uploads)	**0**
Lighthouse mobile: perf. / access.	**85 / 94**
Lighthouse: best practices	**100**
Security: CORS, no-store, upload limits, Firestore rules	**Pass**

Additional Details/Future Development (if any)

What we built, and what comes next.

Built in the prototype

- Explain any document from photos or PDFs in 9 languages
- 23 document kinds recognised; 12 with their own layout
- Multi-document PDFs split into parts
- Evidence labels, dates calculated in code
- Listen, Ask by text or voice, Share, Save as image
- Medicine and deadline reminders; results kept on the phone

Next, after shortlisting

- Faster results in Indian languages
- Compare reports over time (lab trends)
- Push reminders and a native app wrapper
- Odia voice and more languages
- Translations reviewed by native speakers
- Tests with older adults and a DPDP privacy review
- Shared cache for public documents; a lighter model for small tasks

What we chose not to claim

- Sarvam does not give medical, legal or financial advice.
- Translations are made by AI. Check the original for anything important.
- It does not verify that a document is genuine.
- Reach and impact are potential, not measured yet.

Known limits today: a 5-page PDF took 45 s and a photo 71 s in testing; 3 of 99 sample results fall back to English (Bengali and Odia); Odia is read aloud with the phone's own voice; photos give "Check against original" instead of Matched; if the privacy check is busy, Share sends a shorter safe summary; no Content-Security-Policy header yet.

Provide links to your:

Try Sarvam in under two minutes.

GitHub Public Repository

https://github.com/vigneshgeni/sarvam

Scan to open

Demo Video Link (3 Minutes)

https://youtu.be/uju7hKk9GzU

Scan to open

Final Product Link

https://sarvam-510715.web.app

Scan to open

Open the app and tap any of the 11 sample documents. No sign-up. Then upload your own photo or PDF.

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

